A3-ENDS

Forensic Investigation Report

Incident ID: ddf3c1a6
Generated Date: 2026-07-04T10:45:46.026011
Case Status: CLOSED - LOGGED
Prepared By: Autonomous AI System

EXECUTIVE SUMMARY

OVERALL RISK LEVEL	THREAT CLASSIFICATION	PREDICTION CONFIDENCE
UNKNOWN	Unknown	0.0

1. Executive Summary

An intrusion detection event was flagged, but the system did not classify it as an attack or assign any severity, risk score, or confidence level. This lack of categorization suggests the event may be benign, a false positive, or an anomaly that requires further investigation. While the immediate threat appears minimal, it is essential to assess the event thoroughly to rule out any potential risks or gaps in detection capabilities.

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2. Threat Analysis

- ***Event Classification**: The event lacks classification as an attack, indicating uncertainty in its nature.*
- ***Possible Causes**:*
  - *Benign activity misflagged by the detection system.*
  - *A low-priority or non-malicious anomaly.*
  - *A potential gap in detection rules or signatures.*
- ***Detection Gaps**: The absence of severity, risk score, and confidence metrics suggests the detection system may not have sufficient context or data to classify the event accurately.*

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3. Potential Impact

- ***Low Immediate Risk**: Without evidence of malicious intent or impact, the event poses minimal immediate risk to the organization.*
- ***Operational Disruption**: False positives or unclassified events can lead to unnecessary resource allocation and alert fatigue for SOC teams.*
- ***Detection Blind Spots**: If the event represents a new or evolving threat, the lack of classification could indicate a gap in the organization's detection capabilities.*

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4. Recommended Actions

1. ***Investigate the Event**:*
   - *Review raw logs and contextual data associated with the event to determine its nature.*
   - *Correlate with other security tools (e.g., SIEM, EDR) for additional insights.*
2. ***Assess Detection Rules**:*
   - *Evaluate the detection rules or signatures that triggered the event to ensure they are up-to-date and accurate.*
   - *Fine-tune rules to reduce false positives and improve classification accuracy.*
3. ***Monitor for Similar Activity**:*
   - *Track for recurring or similar events to identify patterns or potential threats.*
4. ***Enhance Detection Capabilities**:*
   - *Consider integrating threat intelligence feeds or advanced analytics to improve the system's ability to classify events accurately.*

5. ***Document and Report***:

    - *Document the findings and actions taken for future reference and reporting to stakeholders.*

6. ***Educate SOC Team***:

    - *Provide training to SOC analysts on handling unclassified or ambiguous events effectively.*

By taking these steps, the organization can ensure that unclassified events are thoroughly investigated, detection capabilities are optimized, and potential risks are mitigated.

INCIDENT INFORMATION

SOURCE IP	DEST IP	PROTOCOL	DETECTION TIME
N/A	N/A	N/A	2026-07-04T10:45:46.026011

AI EXPLAINABILITY (SHAP)

TOP CONTRIBUTING FEATURES

SYSTEM RESPONSE

- **Anomaly Detected**
  Sparse Autoencoder MSE: N/A
- **Classification Complete**
  LightGBM assigned label: Unknown
- **Policy Evaluated**
  Severity UNKNOWN processed.
- **Action Executed**

DRIFT & ADAPTATION

CONCEPT DRIFT STATUS

STABLE

ADWIN DETECTION

No significant distribution shift detected in risk scores.

MODEL RETRAINING REQUIRED

False

TECHNICAL EVIDENCE

RISK SCORE	ANALYST OVERRIDE
0 / 100	Pending